

# Eliseu Venites Filho

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## SKILLS

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**Languages:** Portuguese (Native), English (Fluent), French (Advanced)

**Programming:** C, C++ (boost), Rust (tokio), Python (Pandas, NumPy, SciPy, scikit-learn, matplotlib, PyTorch, TensorFlow), Julia (DataFrames.jl, Plots.jl, Makie.jl), Haskell (megaparsec), Delphi

**Tools:** Databases (SQLite, PostgreSQL, DuckDB), Docker

## EXPERIENCE

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### Software Engineer

*Nelógica*

Porto Alegre, Brazil

*Feb. 2026 - Present*

- Engineer low-latency, high-performance backend microservices in Delphi as part of the Automation Tools team, serving thousands of daily active users
- Sustain high availability on mission-critical legacy systems through proactive maintenance and infrastructure hardening
- Reduce manual processing time by automating workflows and building custom internal tools that scale operations across teams
- Deploy and operate services across AWS, Azure, and Equinix infrastructure, using GitLab CI pipelines within an Agile/Scrum team workflow



### Ph.D. in Computational Statistical Physics

*Universidade Federal do Rio Grande do Sul*

Porto Alegre, Brazil

*Jul. 2021 - Feb. 2026*

- Doctoral research funded by a CNPq scholarship
- Analysis of the ensemble correlations of observables of complex systems in order to predict their critical behavior
- Systems from different Universality Classes considered
- Both systems with and without a defined Hamiltonian considered
- The simulations were implemented in Rust while the data analysis was done in the Julia ecosystem



### M.Sc. in Computational Statistical Physics

*Universidade Federal do Rio Grande do Sul*

Porto Alegre, Brazil

*Mar. 2019 - May 2021*

- Master's research funded by a CNPq scholarship
- Scored higher than 99.42% of candidates on the EUF 2-2018 (National graduate programs entrance exam)
- Performance evaluation of the Simulated Annealing applied to different configurations of the Traveling Salesman Problem
- Analysis of the stochastic optimization algorithm applied to problems at the boundary between P and NP complexity classes
- The optimization algorithm was implemented in C++ while the data analysis was done in the Python ecosystem



### Optical Engineering Internship

*Télécom ParisTech*

Paris, France

*Jun. 2017 - Sep. 2017*

- Improved photon-pair coincidence count rate by engineering active stabilization of a polarization-entangled photon-pair source
- Enabled reliable Quantum Key Distribution (QKD) protocol testing by optimizing signal count rates within the Information Quantique et Applications research group



### Undergraduate Research Assistant (Scientific Initiation)

*Universidade Federal do Rio Grande do Sul*

Porto Alegre, Brazil

*Jun. 2013 - Jun. 2015*

- Conducted CAPES-funded undergraduate research in Quantum Information
- Presented original research on Shor's Algorithm for Integer Factorization at the UFRGS XXVI Scientific Initiation Meeting (2014)

## EDUCATION

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**Institut d'Optique Graduate School**  
Diplôme d'Ingénieur

Palaiseau, France  
Sep. 2015 - Sep. 2017

- Double degree in the context of BRAFITEC program
- Optical Instrumentation, Automation, Lasers and Quantum Optics



**Université Paris-Saclay**  
L3 et M1 en Physique Fondamentale

Orsay, France  
Sep. 2015 - Sep. 2017

- Double degree in Theoretical Physics offered to engineering students
- Analytical Mechanics, Statistical Physics, Plasma Physics and Atomic and Molecular Physics



**Universidade Federal do Rio Grande do Sul**  
B.Sc. in Engineering Physics

Porto Alegre, Brazil  
Mar. 2013 - Dec. 2018

- Graduated Summa Cum Laude with a final GPA of 9.54/10.0

## PROJECTS

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### [tsp-sa](#) | C++ / Python

- Built a modular C++ optimization library implementing Simulated Annealing for the M.Sc. research
- Supports Generalized Simulated Annealing and Tsallis Entropy statistics
- Decoupled the optimization core from the TSP implementation, enabling reuse across arbitrary Markov chain problems
- Automated data analysis and plotting in Python to accelerate research iteration

### [artificial-systems](#) | Rust (ndarray, serde)

- Implemented high-performance Rust simulations of computational models for the Ph.D. research
- Modeled Spin Systems (Ising and Blume-Capel models) at scale
- Investigated the Contact Process with diffusion

### [ts-cov-matrix](#) | Julia (DataFrames.jl, Makie.jl)

- Applied Random Matrix Theory to analyze time-series covariance matrices for the Ph.D. research
- Compared spectral properties against the Marchenko-Pastur distribution across NOAA temperature records, Spin Systems, and Contact Processes
- Built a full data analysis pipeline in the Julia ecosystem (DataFrames.jl, Makie.jl)

### [json-parser](#) | Haskell

- Strict JSON parser implemented in Haskell using Megaparsec
- Adheres closely to JSON standards
- Can be used as a library or a standalone command-line tool

### [sternhalma-server](#) | Rust (tokio)

- Asynchronous game server for Sternhalma (Chinese Checkers) built with Rust and Tokio
- Actor-like architecture with decoupled game logic and connection handling
- Client-agnostic design supporting CLI, GUI, and AI agents
- Supports both Raw TCP and WebSocket connections using a CBOR-based protocol

### [sternhalma-agent \(WIP\)](#) | Python (PyTorch)

- Reinforcement learning agent implementing AlphaZero from scratch
- Uses Monte Carlo Tree Search (MCTS) for planning and Deep Neural Networks (ResNet) for evaluation
- Designed to master Sternhalma through self-play without human knowledge

## PUBLICATIONS

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### [Revisiting the Contact Model with Diffusion Beyond the Conventional Methods](#)

2025

*Symmetry*

R. da Silva, E. Venites Filho, H. A. Fernandes, P. F. Gomes

### [Efficient computational method using random matrices describing critical thermodynamics](#)

2025

*International Journal of Modern Physics C*

R. da Silva, E. Venites Filho, S. D. Prado, J. R. Drugowich de Felício

 **A Spectral Investigation of Criticality and Crossover Effects in Two and Three Dimensions: Short Timescales with Small Systems in Minute Random Matrices**

2024

*Entropy*

*E. Venites Filho, R. da Silva, J. R. Drugowich de Felício*

 **Mean-Field Criticality Explained by Random Matrices Theory**

2023

*Brazilian Journal of Physics*

*R. da Silva, H. C. M. Fernandes, E. Venites Filho, S. D. Prado, J. R. Drugowich de Felício*

 **A Thorough Study of the Performance of Simulated Annealing in the Traveling Salesman Problem under Correlated and Long Tailed Spatial Scenarios**

2021

*Physica A: Statistical Mechanics and its Applications*

*R. da Silva, E. Venites Filho, A. Alves*